

AgTrust: Framework for Securing Critical Agriculture Technology and Emerging Solutions

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Abstract—Agriculture technology (AgTech) applies emerging technologies, such as the Internet of Things (IoT), sensors, and artificial intelligence to collect, transmit, and process information related to agricultural enterprises. AgTech solutions involve many interconnected systems, each with a range of security vulnerabilities that may impact data trustworthiness. To engage AgTech effectively, a framework is necessary for providing assurances that data used for decision making can be trusted. This article introduces a novel framework “AgTrust” that provides a cohesive solution for enabling trust in farming ecosystems. This includes assurances of data integrity, data quality, and data provenance supported by anomaly detection, quantum-safe encryption, authentication, and the underlying key-management mechanisms. The AgTrust framework is a modular design, intended for flexibility to enable implementation in a range of context and to meet the challenges of a rapidly evolving threat landscape using appropriate security mechanisms to ensure data trustworthiness. The modules providing various security functions can be adapted to future-proof the AgTech development and can be applied to other IoT or smart-device-based networks. The AgTrust framework specifies the functionality required across various layers and services and coordinates interactions between functions. Four case study examples related to the implementation of the AgTrust framework in real-world scenarios are presented and discussed.

Index Terms—Automated anomaly detection, cyber threats, data trustworthiness, Internet of Things (IoT), postquantum cryptography, security architecture.

I. INTRODUCTION

MODERN farming practices utilize networks of sensors, actuators, and electronic control systems, integrated with various technologies for collecting, storing, transmitting, and processing data associated with control of cyber-physical systems. Agriculture technology (AgTech) applies devices, protocols, and computational paradigms to enable data-informed

agricultural processes for increased productivity in changing climatic conditions. Emerging technologies, such as artificial intelligence, the cloud, and edge computing provide strong capabilities for storing, analyzing, and retaining massive amounts of data (big data) generated by smart farms [1].

A. Research Motivation

In a digital farming scenario, data originates from heterogeneous sources. For example, sensors located in the fields provide data on soil moisture, temperature, and carbon levels to farm systems using technologies, such as Internet of Things (IoT). Farm data can be combined with data obtained from external sources, such as weather forecasting, satellite imagery, and market reports on stock and crop prices. These various data items may have different characteristics in precision, types, ranges, and units, and may have inherent uncertainties. Data collected from these multiple sources is used for making critical farming decisions using different analytical methods, including automated techniques, such as shallow and deep machine learning (ML) [2].

B. Research Challenge

AgTech solutions use data to inform farming practice decisions. The trustworthiness of the data used in these processes has a direct impact on the farming outcomes. However, data are not inherently trustworthy. Poor quality or erroneous data may affect the quantity and quality of farming products [3]. For example, false soil moisture readings may trigger irrigation systems leading to overwatering and subsequent crop damage. This is environmentally and financially damaging [4].

Data trustworthiness is critical for quality data-informed decision-making. Data trustworthiness can be expressed as a combination of the following three properties [5].

- 1) *Data integrity*: Assurance that information has been created, amended, or deleted only by authorized entities [6].
- 2) *Data quality*: High-quality data correctly represents the real-world item it refers to [5].
- 3) *Data provenance*: Being able to track and validate data through the sequence of steps taken in an information flow [5].

C. Existing Research Gap

AgTech solutions involve many interconnected systems, each with a range of security vulnerabilities that may impact data trustworthiness. There are problems obtaining assurances about

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the confidentiality, integrity, and availability of data and authentication across the entire AgTech ecosystem. To engage AgTech effectively, a framework is necessary for providing assurances that data used for decision making can be trusted.

Some IoT architecture models [7], [8], [9], [10], [11] are aligned with a particular vendor's product range. The security components of these frameworks are often part of proprietary products not well-defined and not well explained. The way that security mechanisms should be implemented is not clear. This increase the complexity of coordinating the operations of products from multiple vendors in a secure way.

The AgTrust framework is intended to address this gap by providing more context around the security processes than the following models, in a vendor-agnostic manner.

D. Research Brief and Novelty

The AgTrust framework ensures that the data used to make critical farming decisions is trustworthy and of high quality. Data trustworthiness is constantly assessed, from the point of collection through to transmission, storage, and use. The framework provides a structure that ensures data are validated for trustworthiness at the source, at the edge, and in the cloud. Provisions are also included to ensure data at rest is encrypted to mitigate against potential data breaches.

The AgTrust framework is a modular design. This allows flexibility and the concurrent use of different security mechanisms to ensure data trustworthiness. The modules providing various functions can be adapted to ensure a future-proof design.

E. Research Contribution

Our AgTrust framework provides a cohesive solution for farming ecosystems, including assurances of data integrity, data quality, and data provenance supported by anomaly detection and quantum-safe (QS) encryption, authentication, and the underlying key-management mechanisms. Effective AgTech incorporating AgTrust will ensure resilient farming systems. The AgTrust framework addresses the following four objectives.

- 1) Validating that the source of the data is authentic, and the data are representative of the real world.
- 2) Ensuring that the confidentiality and integrity of the data is maintained from the point of collection to point of use (smart devices to the edge, and from the edge to the cloud).
- 3) Maintaining appropriate level of confidentiality and integrity when the data is at rest.
- 4) Preventing the disclosure of data during processing to maintain data confidentiality.

F. Organization of This Article

The rest of this article is organized as follows. Section II defines an envisioned system and the associated information flow to demonstrate AgTech systems of differing complexities. Section III gives an overview of security in existing IoT reference architectures. Section IV provides our proposed AgTrust framework. Section V discusses data flow between the modules and components of the AgTrust framework in three stages

of communication. Section VI presents case study examples related to the implementation of the AgTrust framework in four real-world scenarios; the functionality/capability is clearly demonstrated. Section VII comments on the QS implementation of AgTrust and discusses the high-entropy cryptographic key generation/distribution. In Section VIII, we discuss the advantages of our proposed framework compared to current other methods. Finally, Section IX concludes this article.

II. ENVISIONED SYSTEM AND INFORMATION FLOW

With the adoption of digital agriculture, the flow of information is not always direct, and often involves complex systems. Information is collected by sensors and transmitted across networks that may not be entirely contained within the farm. Multiple parties may have access to this information at different points.

In this section, we associate the AgTech products with three different sensors: pressure, soil moisture, and livestock weight (refer to Fig. 1). Note that these scenarios are used to demonstrate systems of differing complexities. It is not intended to imply that the information will always flow in this way with these AgTech solutions.

The *purple* information flow begins with data collected from a soil moisture sensor. The information from the sensor is transmitted to a wireless access point on the farm via a Wi-Fi connection. From the access point, the data are sent to a local server on the farm for processing. The farmer can access the information using their laptop, which is also connected via Wi-Fi to the wireless access point. In this information flow, *the data remain on the farm*.

The *red* information flow begins with data collected from a pressure sensor. The information from the sensor is transmitted to a tower via the cellular 4G network. From the cellular tower, the data are sent to *a server outside the farm* for processing. The farmer can access the information using their laptop, which is connected via Wi-Fi to the wireless access point. In this information flow, the pathways from the cellular tower to the external server and from that server to the farmer are uncertain. This is a more complex setup than the purple information flow, and the farmer has less control of their data. The location, ownership, and security practices of the various external systems and networks the data passes through may be unknown. There are multiple places in which data trustworthiness may be impacted.

The *blue* information flow begins with data collected from a livestock weight scale. The information from the scale is transmitted to the farmer's smartphone via a direct bluetooth connection. The information is then accessed by the farmer using an application on the smartphone. The mobile application then transmits the data to *a server external to the farm* for storage and processing via the local Wi-Fi access point. Again, this information flow is more complex than the purple information flow and the farmer does not have full control of their data. The location, ownership, and security practices of the various external systems and networks the data passes through may be unknown. There are multiple places in which data trustworthiness may be impacted.

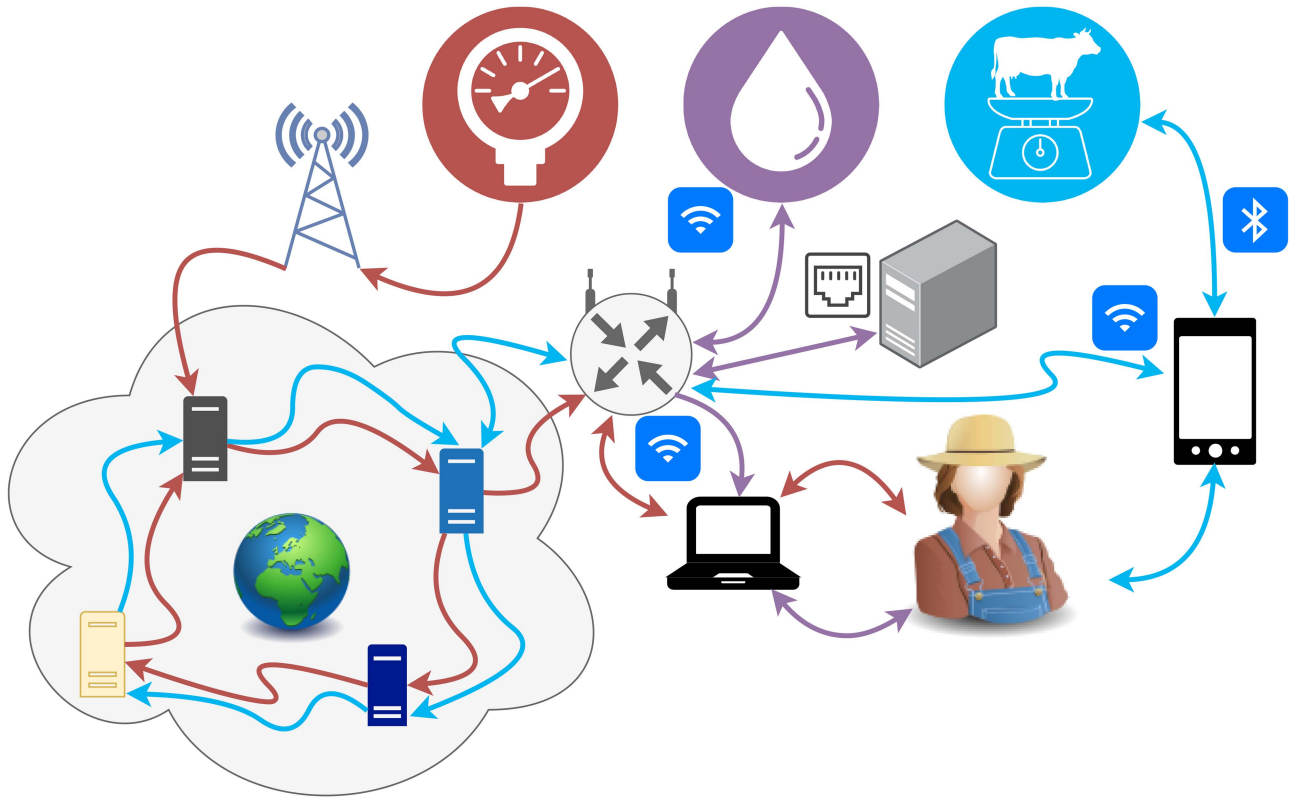


Fig. 1. Envisioned system model and three examples of information flow in the smart farm environment.

To accurately assess risks to data trustworthiness, awareness of all the stages of data flow from data collection, transmission, processing, and storage is important [12]. If risk is assessed based on assumptions that the information flow is direct (similar to the *purple* information flow) when it is actually following a more complex path (similar to the *red* or *blue* information flows), the threats to data trustworthiness are not being adequately assessed. The range of threats to be considered depends on the level of complexity of the information flow, and could involve components (e.g., sensors, network attached storage devices, vendor cloud-based systems) used for data gathering, storage, and transmission both on and off the farm.

III. RELATED WORK

The Intel IoT system architecture [7] is designed to provide a framework to connect most devices to cloud based systems, even if the devices are not IP-based. The architecture describes how different components including “things” end devices, “network” infrastructure, and “cloud” computing modules interact. Intel reinforces the need for security at all endpoints, network infrastructure, and cloud components. However, the exact security mechanisms that should be implemented are not clearly specified.

The Cisco IoT security reference model [8] presents aspects of security that should be considered in IoT, and indicates the place in the reference model where these aspects are relevant. This model shares similarities with the well-known and widely used

seven-layer OSI model: defining the lower level functions at the bottom, such as hardware devices; with application, software, and processes appearing in the higher levels. The Cisco model specifies that security should be present at each of Cisco’s defined seven levels. It specifically mentions that security must be applied at a physical level, network level, and through the software levels. While it identifies key areas, such as authentication, authorization, secure protocols, and secure storage; how it achieves these goals is unclear.

The Microsoft Azure IoT reference architecture [9] recommends technology choices and implementation approaches specifically for Azure products. The model describes the data flow in a one way flow beginning with things (IoT devices) that generate data. The data then flow to Insights, which are generated from the data. The data flow concludes with actions, which can be taken based upon the Insights. Microsoft [9] shows the “all subsystems and cross-cutting needs” model, highlighting all the core components of the architecture, together with other considerations that should be implemented, including: security, logging, and monitoring, and high availability and disaster recovery. The model recognizes that security considerations are critical.

The IBM Watson IoT architecture [10], [11] is another vendor-specific solution. It makes use of the IBM Watson IoT Platform and illustrates the data flow through the architecture within different steps, including: data collection, converting traffic from operational technology protocols to IP-based communication at the gateway, data filtering and aggregation, data analysis, data output, and data storage.

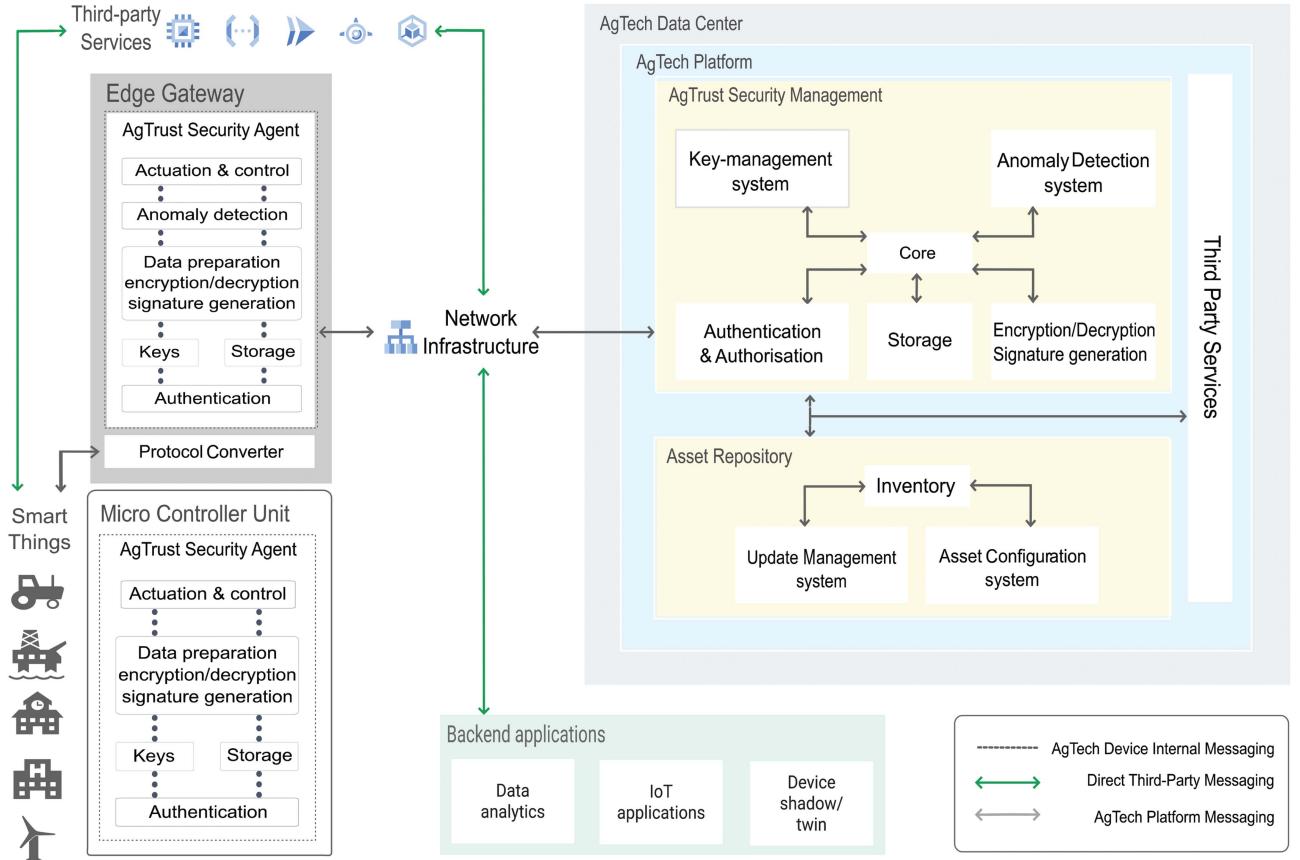


Fig. 2. Our AgTrust framework.

IV. AGTRUST FRAMEWORK

Fig. 2 illustrates our AgTrust framework and its components. The shaded section in the framework marked as data center and houses the AgTrust security platform and other security-related and nonrelated systems. The AgTech data center may be a local data center within the digital farm, a cloud-based data center, or exist in a hybrid form.

A. AgTech Platform

The AgTech platform is a container to logically group together the AgTrust security management system (the heart of the AgTech platform) and other systems, such as third-party services and asset repository that are external to the main functionality of the framework that are relevant and/or dependent on the security management system. Asset repository is related to the asset management technology. All IoT devices used within the farm should be provisioned into the asset repository.

B. Asset Repository

This system is related to the asset management technology. All IoT devices used within the farm should be provisioned into the asset repository. The information stored within the asset repository assists the AgTrust security management system in identifying legitimate network devices.

C. AgTrust Security Management

The AgTrust security management system uses the information stored within the asset repository to assist in identifying legitimate network devices. If a device is not provisioned in the asset repository and attempts communication, these communications will be blocked by the security management system. Below, each module of the AgTrust security management system is defined.

1) **Core:** This is the parent module of all other modules within the AgTrust security management system. The core is responsible for coordinating all activities of the modules. The other modules should not interact directly. All communication between modules is via the core. Once an entity has been authenticated and authorized, the transmission is passed to the core to determine the next action that must be taken.

2) **Authentication and Authorization:** This module is responsible for ensuring the edge gateways are authenticated (entity authentication) to the AgTrust system, and that the data has not been changed in transit (message authentication). It is also responsible for ensuring that effective access control measures are implemented to ensure that only authorized devices can read/write data. This module is also responsible for ensuring secure network communications from any external systems and data sources. All communications must pass through this module before being passed to the core module.

3) *Key Management System*: This module acts as the root of trust (certificate authority) in the AgTrust framework. This module is responsible for generation, distribution, revocation, destruction, and update of the cryptographic keys [13] that are used by all participant devices including the sensors and the edge. This module is responsible for generating keys of an appropriate length. A hardware security module should be used to ensure high entropy randomness. The keys that are generated must be QS and should comply with the latest recommendations of standardization bodies, such as NIST [14].

4) *Anomaly Detection System*: This module is responsible for the ML related tasks that must be completed to ensure that anomalous sensor values, or outliers [15], [16] are detected and processed. The primary purpose of this module is to ensure that the sensor values are of sufficient quality before being stored or used in other applications. The anomaly detection module is distributed, meaning that basic bounds checking can be applied directly on the sensor node, a basic level of ML tasks can be completed at the edge. More complex ML tasks can be handled by the AgTrust platform in the data center. The training of the ML models should take place in this module in the AgTrust platform. The trained models can then be distributed to the edge gateways, and where possible the sensor nodes.

5) *Storage*: This module is responsible for ensuring that all data are stored securely. Data at rest must have appropriate encryption applied. Encryption of data at rest ensures confidentiality is maintained in the event of a breach.

6) *Encryption/Decryption Signature Generation*: This module is responsible for ensuring that transmissions are signed with an appropriate digital signature to ensure nonrepudiation. This module is also responsible for the encryption of outgoing transmissions, and the decryption of received transmissions. For a more detailed explanation of these functions and components.

D. Backend Applications

The backend applications section of the framework diagram represents the multitude of cloud applications and platforms that may be used by the end user to access their data. These applications are external to the AgTrust platform. The data ideally should have been processed by the AgTrust platform before it is accessed by the end user by way of an API provided by the application. In cases where this is not possible, data should be exported from the application to the AgTrust platform for anomaly detection purposes [15], [16].

E. Smart Things

The smart things section represents the numerous smart devices, sensors, and actuators that are deployed within the farm. The devices in this section may or may not be capable of direct communication with the edge gateway and the AgTrust security agent. Certain proprietary vendor supplied solutions may communicate with third party services directly via communication technologies, such as cellular.

F. Third-Party Services

The third-party services section represents third-party vendor provided platforms that proprietary solutions may communicate with that are outside the control of the AgTrust platform. The AgTrust platform may ideally obtain data from these solutions from vendor supplied APIs. Alternatively, data may be exported from these platforms and imported to the AgTrust platform through other more manual processes where no other options exist.

G. Edge Gateway

This section represents the gateways that are responsible for communicating with the various devices such as sensors and actuators that are deployed in the farm. The edge gateway contains two modules—the AgTrust security agent, and a protocol converter.

1) *Protocol Converter*: The protocol converter module is responsible for obtaining data from multiple sensor devices. Many of the sensors may not be IP based, using various technologies such as Bluetooth, Zigbee, and LoRa [17]. This module is responsible for ingesting this network traffic and converting it to IP based traffic so that it can be processed by the AgTrust security agent. This module represents the varying hardware components and software components that could be involved in this conversion process.

2) *AgTrust Security Agent*: The AgTrust security agent acts as a bridge between the smart things/sensors and the AgTrust security management platform. The sensors communicate with the AgTrust security agent, rather than directly with the AgTrust security management platform. In this article, each module of the AgTrust security agent is defined as follows.

Authentication module is responsible for ensuring all directly connected smart devices have been authenticated. The security agent must provide assurance to the AgTrust security management platform that all devices under its responsibility are authenticated.

Keys module is responsible for the storage and management of the various keys that the edge gateway needs to communicate with the sensors, and with the AgTrust security management platform.

Storage module is responsible for the secure storage of locally relevant data before it is moved to the AgTrust security management platform.

Data preparation, encryption/decryption, and signature generation module is responsible for ensuring that transmissions are correctly formatted, and signed with an appropriate digital signature to ensure nonrepudiation. This module is also responsible for the encryption of outgoing transmissions, and the decryption of received transmissions.

Anomaly detection module is responsible for anomaly and outlier detection of the sensor data. Where possible, anomaly detection will take place on the edge gateway. The rules that are used by this module are defined by the anomaly detection system module of the AgTrust security management system.

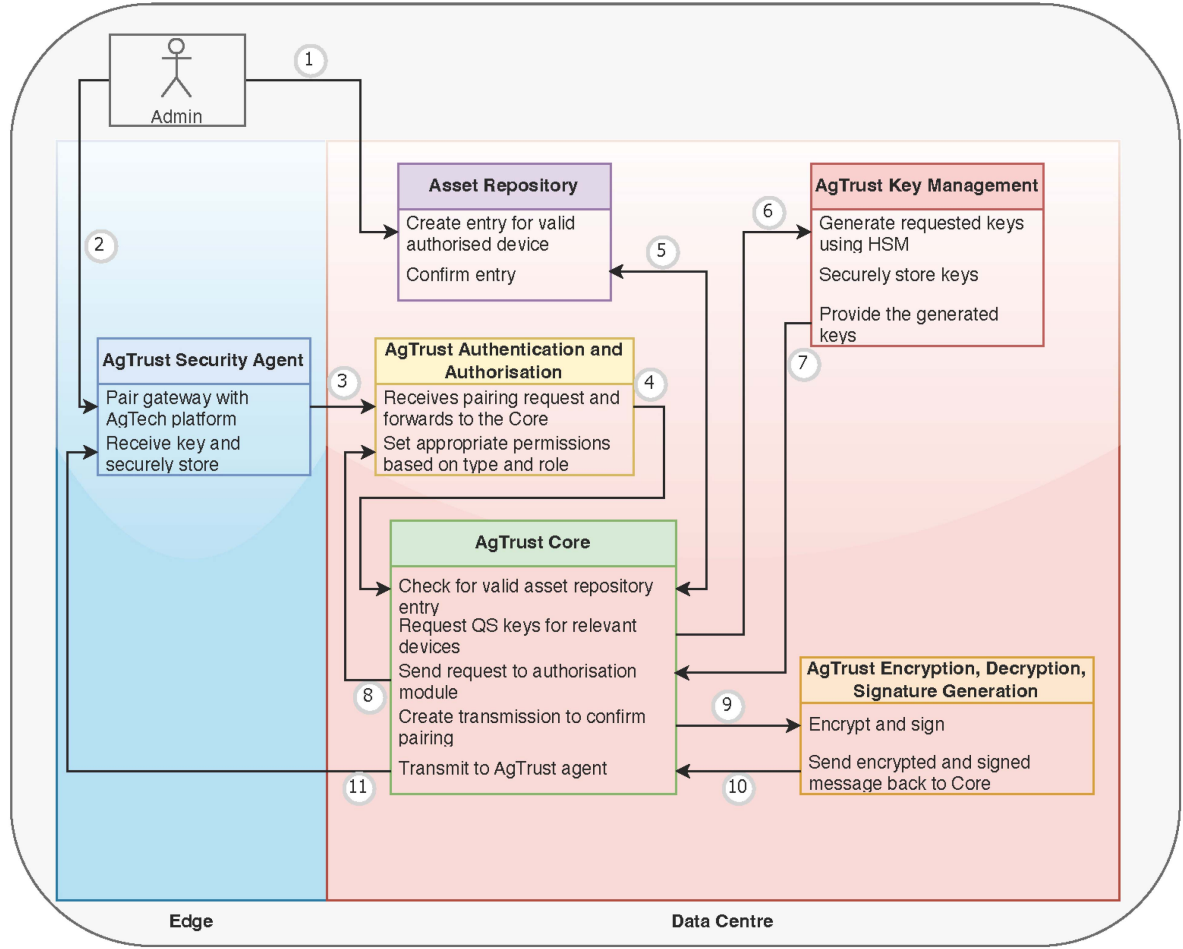


Fig. 3. Data flow required in provisioning an edge gateway device with AgTrust.

Actuation and control module is responsible for initiating management functions on devices to enable control, and also for issuing instructions to operate any actuators.

H. Security Agent in Smart Things

All the components that make up the smart devices on the farm, such as sensors, actuators, and associated microcontrollers should be running a local version of the AgTrust security agent.

I. Network Infrastructure

Network infrastructure section represents that varying network infrastructure that the network traffic may pass through. This can include devices, such as switches, routers, load balancers, and firewalls. The exact infrastructure that may be present will vary between deployments. The network infrastructure is not managed by the AgTrust platform.

V. DATA FLOW DEMONSTRATION IN AGTRUST

This section presents data flow between the modules and components of the AgTrust framework in three stages of communication.

A. Data Flow in Provisioning Stage

Provisioning new devices securely using AgTrust is the first step to ensure ongoing secure communication. Provisioning generally only needs to be conducted during the initial installation of a device. There are the following two main types of components that need to be provisioned.

- 1) *Edge gateways*: These devices need to be provisioned to the network securely. These devices are responsible for managing sensors that are directly connected to them.
- 2) *Sensor nodes*: These are the end devices such as sensors and actuators that are deployed in the farm. These devices must also be provisioned and attached to the edge gateway.

The diagram in Fig. 3 shows how the AgTrust security agent in the edge gateway interacts with the various AgTrust security modules in the data center. The diagram is numbered in the order that the interactions take place. Specifically, the interactions that occur in this data flow are as follows.

- 1) The administrator must create an entry for the new device in the asset repository. This must include device identifying information, such as serial numbers, make, model, and function of the device.
- 2) The administrator initiates a provisioning request on the new device.

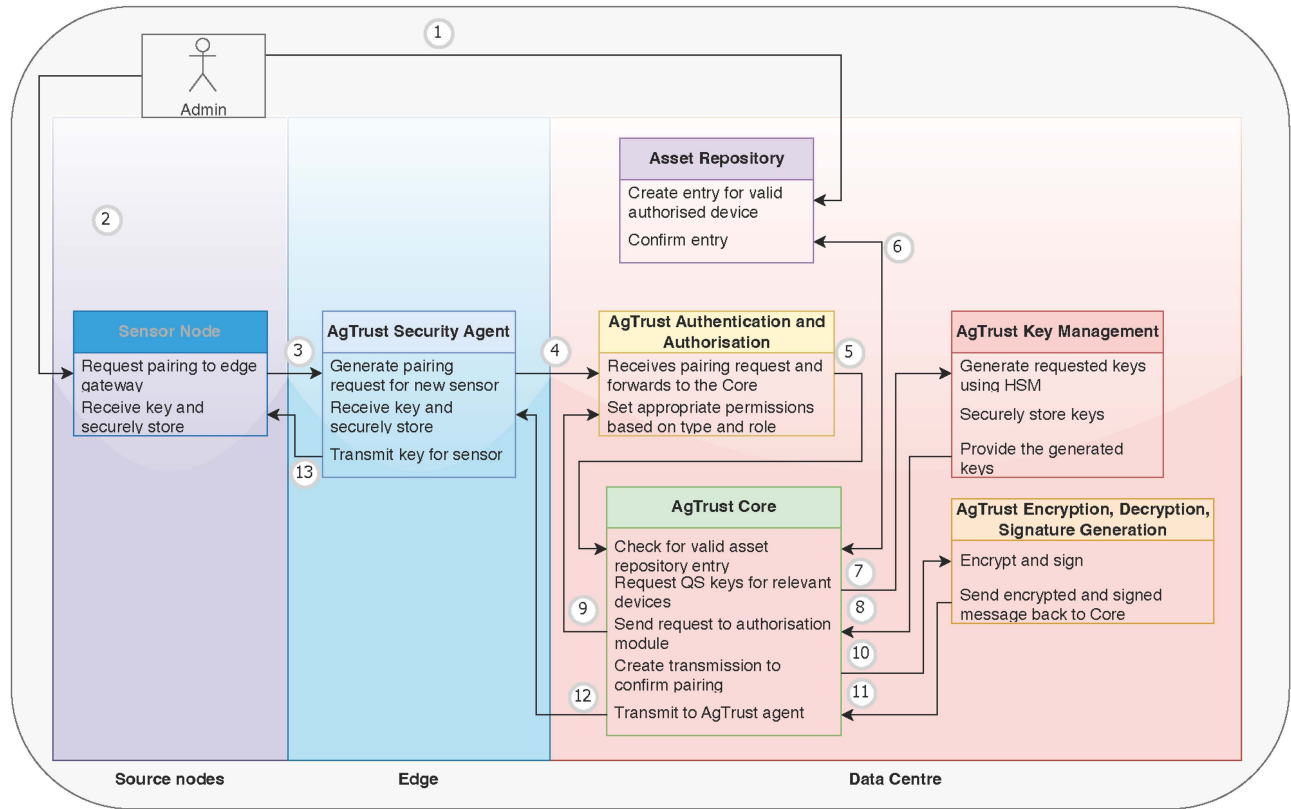


Fig. 4. Data flow required in provisioning a sensor device with AgTrust.

- 3) The edge gateway sends a provisioning request to the AgTrust authentication and authorization module.
- 4) The AgTrust authentication and authorization module forwards the pairing request to the core for action.
- 5) The AgTrust core module confirms that the edge gateway has a valid entry in the asset repository.
- 6) The AgTrust core module requests all the required certificates and keys for the new device to be created by the AgTrust key management module. The AgTrust key management module generates the required keys and certificates with sufficient randomness. The keys are saved and securely stored within the AgTrust key management module.
- 7) The AgTrust key management module provides the required keys and certificates to the AgTrust core module.
- 8) The AgTrust core module requests the creation of appropriate permissions for the new device. These are defined in the AgTrust authentication and authorization module.
- 9) A transmission is created to confirm the pairing request, it is sent to the AgTrust encryption, decryption, signature generation module to be encrypted and signed.
- 10) The encrypted and signed message is sent back to the AgTrust core module.
- 11) The AgTrust core transmits the confirmation message to the AgTrust security agent in the edge gateway, which stores the keys and certificates. This confirms and finalizes the provisioning process.

Fig. 4 illustrates the data flow associated with provisioning a new sensor node. The diagram shows how the sensor node interacts with the AgTrust security agent in the edge gateway, and how the edge gateway interacts with the various AgTrust security modules in the data center. The diagram is numbered in the order that the interactions take place. Specifically, the interactions that occur in this data flow are and follows.

- 1) The administrator must create an entry for the new device in the asset repository. This must include device identifying information, such as serial numbers, make, model, and function of the device.
- 2) The administrator initiates a provisioning request on the new device.
- 3) The sensor node sends a provisioning request to the AgTrust security agent inside the edge gateway.
- 4) The edge gateway generates and forwards the provisioning request to the AgTrust authentication and authorization module.
- 5) The AgTrust authentication and authorization module forwards the pairing request to the core for action.
- 6) The AgTrust core module confirms that the sensor node has a valid entry in the asset repository.
- 7) The AgTrust core module requests all the required certificates and keys for the new device to be created by the AgTrust key management module. The AgTrust key management module generates the required keys and certificates with sufficient randomness. The keys are saved

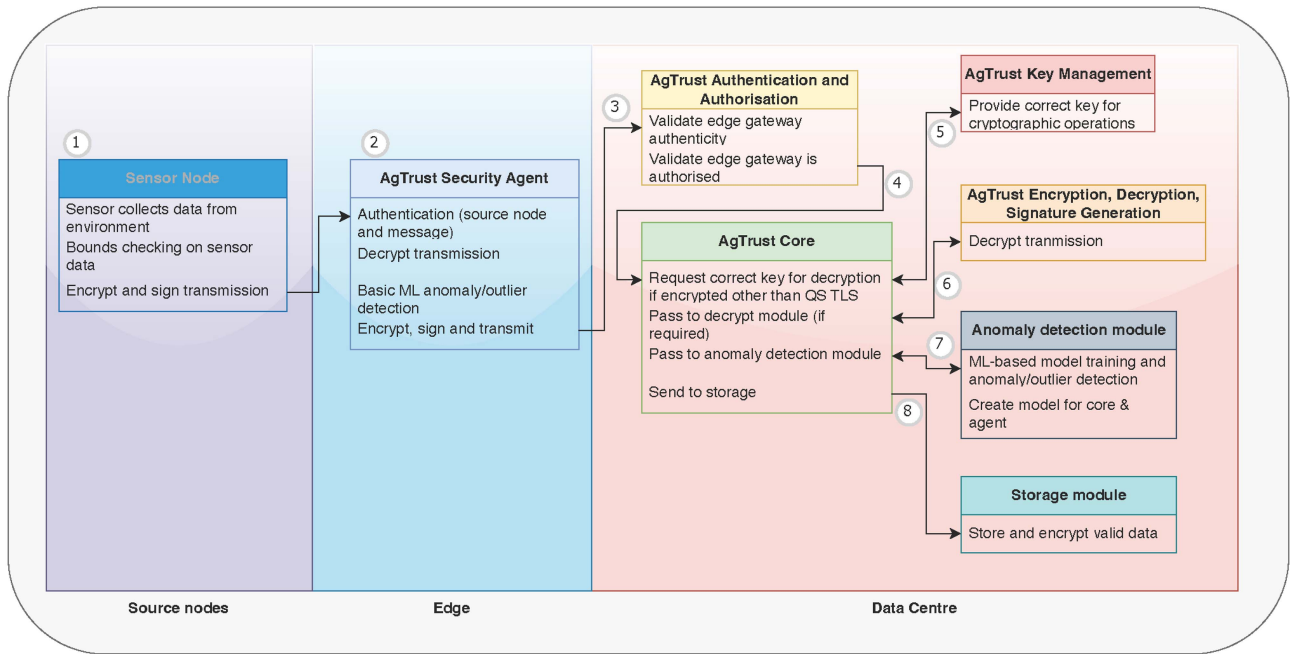


Fig. 5. Data flow required for transmitting sensor readings with AgTrust.

and securely stored within the AgTrust key management module.

- 8) The AgTrust key management module provides the required keys and certificates to the AgTrust core module.
- 9) The AgTrust core module requests the creation of appropriate permissions for the new device. These are defined in the AgTrust authentication and authorization module.
- 10) A transmission is created to confirm the pairing request, it is sent to the AgTrust encryption, decryption, signature generation module to be encrypted and signed.
- 11) The encrypted and signed message is sent back to the AgTrust core module.
- 12) The AgTrust core transmits the confirmation message to the AgTrust security agent in the edge gateway, which stores the keys and certificates.
- 13) The AgTrust security agent transmits the key and certificate to the sensor node, which receives and stores the information. This confirms and finalizes the provisioning process.

B. Typical Sensor Data Flow

Fig. 5 illustrates the data flow associated with the collection of data by a sensor node. The diagram shows how the sensor node interacts with the AgTrust security agent in the edge gateway, and how the edge gateway interacts with the various AgTrust security modules in the data center. The diagram is numbered in the order that the interactions take place. Specifically, the interactions that occur in this data flow are as follows.

- 1) The sensor node collects data from the environment, it performs initial bounds checking on the sensor data. If capable it may also perform more advanced ML based data

quality checking if appropriate. The data are encrypted and signed and sent to the AgTrust security agent in the edge gateway.

- 2) The AgTrust security agent examines the transmission and checks the message and source authenticity. It then decrypts the communication and performs ML-based anomaly detection based on models that were previously provided by the AgTrust security platform. It then encrypts, signs, and transmits the sensor reading to the AgTrust authentication and authorization module.
- 3) The AgTrust authentication and authorization module validates the authenticity of the message, and the source. It also confirms the edge gateway is authorized to provide this particular data.
- 4) The transmission is then forwarded to the AgTrust core module.
- 5) The AgTrust core module receives the transmission and requests the correct key for decryption from the AgTrust key management module. This only occurs if the message was sent by mechanisms other than QS transport layer security (TLS) [18], [19].
- 6) If QS TLS was not used, the data are passed to the AgTrust encryption, decryption, signature generation module for decryption.
- 7) The data are then passed from the AgTrust core module to the anomaly detection module for ML-based anomaly/outlier detection. These data contribute to the training of the models, and the creation of new models if required.
- 8) The data are then sent from the AgTrust core module to the storage module to be stored securely. This confirms and finalizes the data collection process.

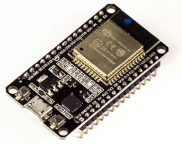


Fig. 6. ESP32 microcontroller is the basis for the custom moisture sensor.

VI. CASE STUDY IMPLEMENTATION

In this section, case study examples related to the implementation of the AgTrust framework in real-world scenarios are discussed and presented. These example scenarios present the data flow, along with the required network infrastructure. The four case studies are as follows.

- 1) A soil moisture sensor that utilizes Wi-Fi network for connection. This sensor is built using nonproprietary hardware and software components.
- 2) A sensor that measures water pressure. This sensor is a proprietary product provided by a vendor. It communicates to the vendor using a 4G cellular connection.
- 3) A scale used to measure the weight of livestock. The scale is a proprietary product provided by a vendor. The scale utilizes a Bluetooth connection to communicate with the vendor's backend system via an application installed on a smart device, such as a phone and tablet.
- 4) A soil moisture sensor that utilizes a Bluetooth network for connection. This sensor is built using nonproprietary hardware and software components. The microcontroller used is heavily resource-constrained with limited processing power and other system resources.

In our case studies, we assume two classes of sensors for demonstration. The first class includes low-powered sensors that are incapable of running public key infrastructure (PKI) [20]. The second class includes more advanced sensor nodes that are powerful enough to run quantum-safe PKI algorithms for key exchange and digital signature generation/verification. This PKI function can be presented between the sensors, the edge device, and the AgTrust security management platform's authentication module.

A quantum-safe Hash-based message authentication code [21] can be used for message authentication between low-powered sensors and the edge device. PKI capable devices can use a quantum-safe digital signature algorithm for message/entity authentication (e.g., CRYSTALS-Dilithium lattice-based signature scheme [22]). Confidentiality of data in transit can be guaranteed by symmetric-key encryption (e.g., AES-256 [23]). Each PKI capable device can use a lattice-based key encapsulation mechanism (e.g., CRYSTALS-Kyber [24]) to establish a shared secret key with other entities. Low-powered sensors can request a new key from the edge. Alternatively, a key rotation function could be applied.

For automated anomaly detection, we aim to detect anything that deviates from the normal pattern "norm." We can use a supervised approach; obtaining datasets of normal sensor readings from a digital farm, but unlabeled. That is, there is no one-to-one correspondence between samples and the associated



Fig. 7. Sensecap S-Temp&VWC&EC-02B soil moisture, temperature, and EC sensor.

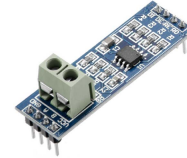


Fig. 8. MAX485 TTL UART to RS485 converter module.

input features, and their corresponding output labels. We can train an autoencoder deep learning neural network model to identify anomalies (exceptions to the norm) within the sensor reading. The effectiveness of this approach is demonstrated in Baee et al.'s work [15], [16].

A. Scenario 1: AgTrust With Soil Moisture Sensor Node Based on Nonproprietary Hardware and Software

This implementation case study demonstrates an example of how the AgTrust framework can be implemented when the developer has full control of the sensor nodes, including the ability to install custom software and firmware. This example presents a soil moisture, temperature, and electric current sensor that is connected to the farm network using Wi-Fi. This section outlines the required hardware and software components. It also presents a diagram that illustrates the data flow that a sensor reading takes using the AgTrust framework throughout its lifecycle, from collection, to storage and use.

The sensor node is based around the ESP32 microcontroller unit (MCU) [25] as seen in Fig. 6. This microcontroller supports Wi-Fi connections and dual mode Bluetooth. The ESP32 is connected to a Sensecap S-Temp&VWC&EC-02B soil moisture, temperature, and electrical current (EC) sensor [26]. The sensor can be seen in Fig. 7. The sensor uses the MODBUS-RTU RS485 protocol [27] to communicate with the MCU. This requires translation as the ESP32 cannot directly connect to the sensor. The sensor is connected to the ESP32 via a MAX485 TTL UART to RS485 converter module as seen in Fig. 8. A diagram of the assembled sensor node can be seen in Fig. 9.

The sensor nodes are connected to the farm Wi-Fi network and have been previously paired to a gateway device that is running the AgTrust security agent software. In the example scenario, this agent is running on a machine that is connected to the gateway device via ethernet. The agent could potentially be installed directly on the access point if resources allow, however, it may be more practical to install this on a dedicated device. This is due to the resource constraints of many network infrastructure devices. These devices commonly have low powered CPUs with limited RAM and ROM. A diagram illustrating these two options is presented in Fig. 10.

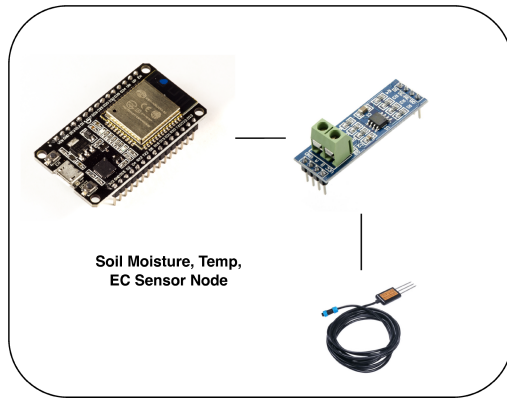


Fig. 9. Assembled sensor node.

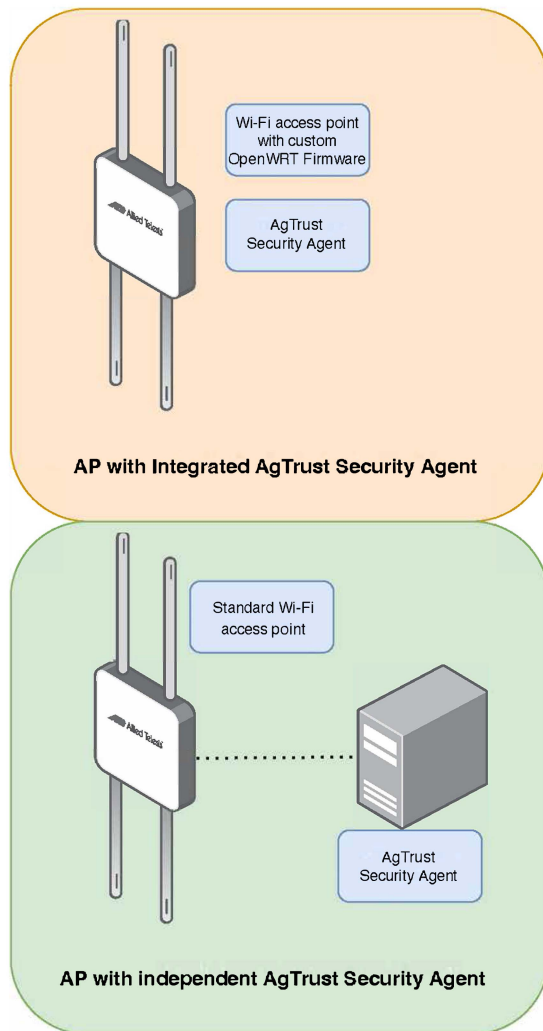


Fig. 10. AgTrust security agent deployment examples.

As this case study example is based on fully open hardware and software, quantum-safe TLS is used to secure the network traffic between the sensor and edge, and the edge and cloud [19]. Fig. 11 shows the hardware required including the sensor node, the edge gateway, network infrastructure, such as switches and routers, as well as the AgTech platform deployed in the cloud.

The network connections are represented by the orange dashed lines. The purple arrows represent the data flow.

The case studies assume that sensor node pairing and gateway device pairing has already taken place. The data flow begins with the sensor node collecting measurements from the environment. The sensor node establishes a connection to the edge gateway via a Wi-Fi connection. A quantum-safe TLS session is established with the edge gateway. The sensor readings are then transmitted to the edge gateway after bounds checking is conducted on the collected values.

The AgTrust security agent ensures that the sensor node is authentic by ensuring that the certificate is valid. The data transmission is decrypted, and anomaly detection processes are applied to the sensor data using previously trained ML models. The data are then encrypted by AgTrust security agent and forwarded via the network infrastructure to the AgTrust security management system within the AgTech platform. The network links are represented by the orange dashed lines, and the data flow is represented by the purple arrows. Each stage of the data flow in Fig. 11 within the AgTrust security management system is numbered.

- 1) The transmission is received by the authentication and authorization module from the edge gateway. The authentication and authorization module ensures that the gateway is authentic, and has the appropriate authorization to provide the sensor data. It also ensures that the transmission is authentic.
- 2) Once the authentication and authorization module is satisfied of the authenticity of the sender and the message, the data are forwarded to the core module.
- 3) The transmission is received by the core module from the authentication and authorization module. The core is responsible for managing and coordinating all functionality related to the AgTrust security management system. The data are then sent from the core module to the anomaly detection module. Anomaly detection processes are applied to the sensor data using previously trained ML models. These data are also used as input to further improve and train the models. Once the anomaly detection module is satisfied that the data is trustworthy, it is forwarded back to the core module.
- 4) The core module then forwards the data to the data storage module to be encrypted and saved.
- 5) User applications can now access the data through secure APIs.

B. Scenario 2: AgTrust With Proprietary Pressure Sensor Node

This implementation case study presents an example of how the AgTrust framework can be implemented with proprietary sensor nodes, and may not have any control over the software and firmware of the end devices. This raises challenges and creates limitations in how the AgTrust framework can operate. The developer does not have access to the security mechanisms of the sensor, and the associated vendor backend systems. The main role of the AgTrust framework in this sort of scenario is to

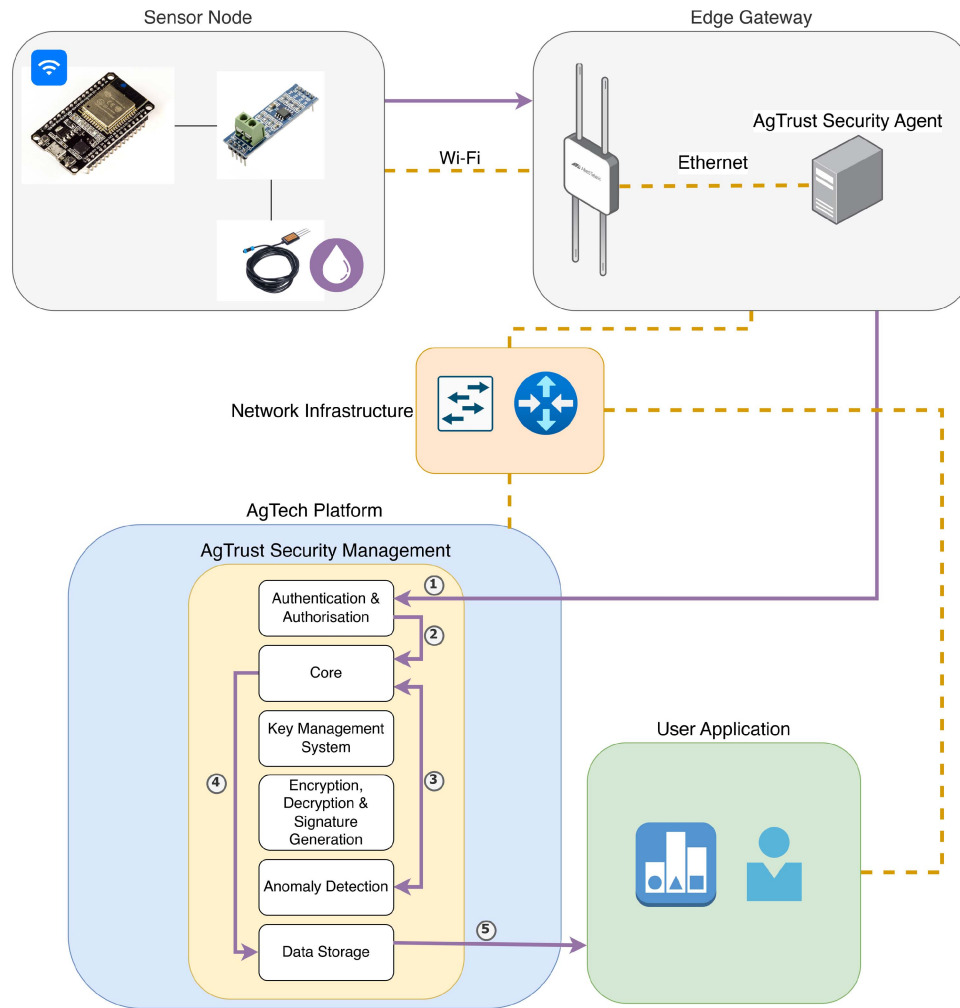


Fig. 11. Case study 1: soil moisture, temperature, and EC sensor AgTrust data flow.

validate the trustworthiness of the data using the ML models, as the AgTrust security management system does not have control or access to the cryptographic processes and systems that the vendor has chosen to implement. The data must be exported from the proprietary system, and then imported into the AgTrust security management system.

In this implementation case study, the sensor node is a Farmbot pressure sensor [28] designed to monitor the pressure of a water line in the farm. These devices can use satellite or cellular technologies to communicate with the vendor's systems. The user is able to access the readings through the vendor's cloud platform. The vendor offers support for integration into a number of existing farm management systems.

Fig. 12 shows the network infrastructure required and the data flow that is associated with the AgTrust framework. Each stage of the data flow is numbered as follows.

- 1) The transmission is received from via export from the farm management system to the AgTrust security management system. It is first processed by the authentication and authorization module. The module ensures that the farm

management system is authentic, and is authorized to provide data from the sensor.

- 2) Once the authentication and authorization module is satisfied of the authenticity of the sender and the message, the data are forwarded to the core module.
- 3) The transmission is received by the core module from the authentication and authorization module. The core is responsible for managing and coordinating all functionality related to the AgTrust security management system. The data are then sent from the core module to the anomaly detection module. Anomaly detection processes are applied to the sensor data using previously trained ML models. These data are also used as input to further improve and train the models. Once the anomaly detection module is satisfied that the data is trustworthy, it is forwarded back to the core module.
- 4) The core module then forwards the data to the data storage module to be encrypted and saved.
- 5) User applications can now access the data through secure APIs.

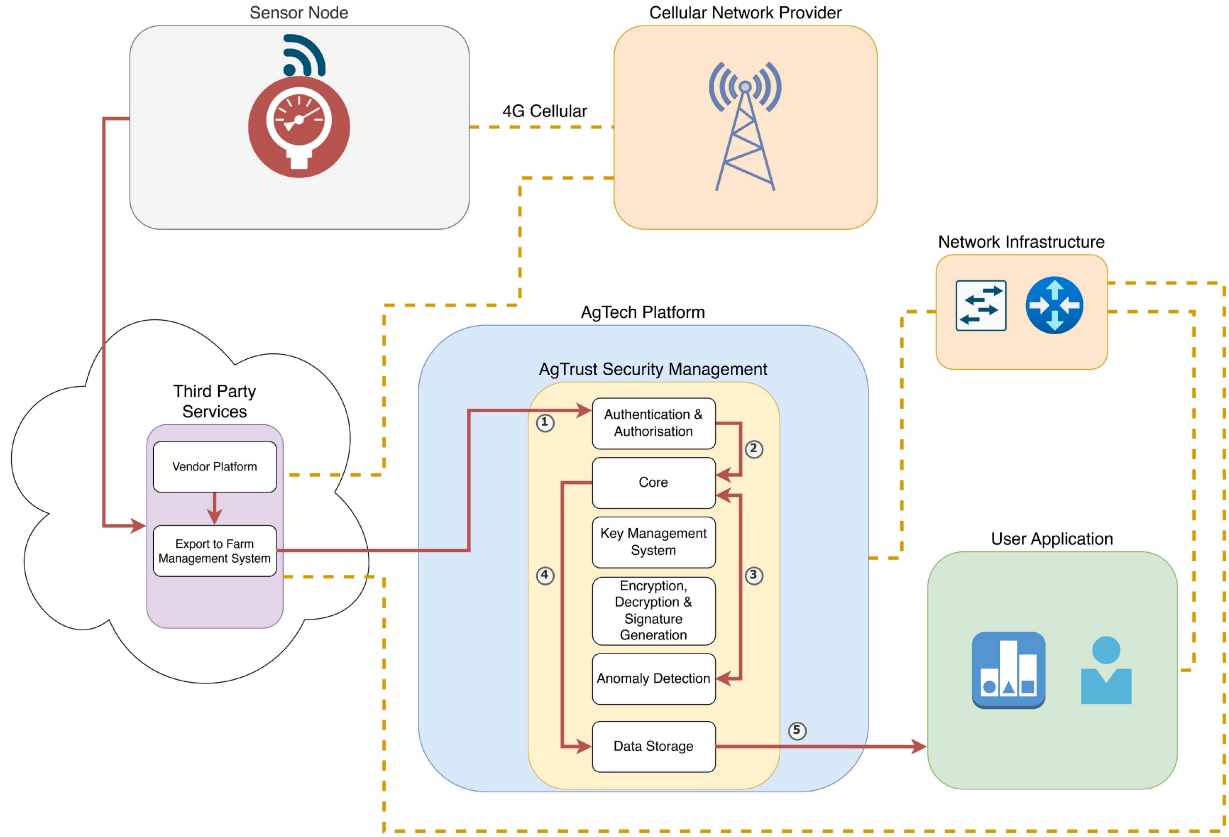


Fig. 12. Case study 2: proprietary pressure sensor AgTrust data flow.

C. Scenario 3: AgTrust With Proprietary Livestock Weight Sensor Node

This implementation case study presents an example of how the AgTrust framework can be implemented with proprietary sensor nodes, and may not have any control over the software and firmware of the end devices. Similar to Case Study 2, this raises challenges and creates limitations in how the AgTrust framework can operate. The developer does not have access to the security mechanisms of the sensor, and the associated vendor backend systems. Again, the main role of the AgTrust framework in this sort of scenario is to validate the trustworthiness of the data using the ML models, as the AgTrust security management system does not have control or access to the cryptographic processes and systems that the vendor has chosen to implement. The data must be exported from the proprietary system, and then imported into the AgTrust security management system.

In this implementation case study, the sensor node is a Tru-Test S3 weight scale indicator that is used to capture the weight of livestock. These devices are designed to connect using Bluetooth to a smart device. The smart device can then export the readings to a range of farm management systems. There is also functionality offered to email the readings for import into other systems.

Fig. 13 shows the network infrastructure required. Network links are represented by the orange dashed line. The data flow that is associated with the AgTrust framework is represented by

the blue arrows. Each stage of the data flow is numbered, as follows.

- 1) The transmission is received from via export from the farm management system to the AgTrust security management system. Alternatively the data are sent by email to the AgTrust security management system. It is first processed by the authentication and authorization module. The module ensures that the farm management system is authentic, and is authorized to provide data from the sensor. Alternatively it ensures that the email received is from an authentic source.
- 2) Once the authentication and authorization module is satisfied of the authenticity of the sender and the message, the data are forwarded to the core module.
- 3) The transmission is received by the core module from the authentication and authorization module. The core is responsible for managing and coordinating all functionality related to the AgTrust security management system. The data are then sent from the core module to the anomaly detection module. Anomaly detection processes are applied to the sensor data using previously trained ML models. These data are also used as input to further improve and train the models. Once the anomaly detection module is satisfied that the data are trustworthy, it is forwarded back to the core module.
- 4) The core module then forwards the data to the data storage module to be encrypted and saved.

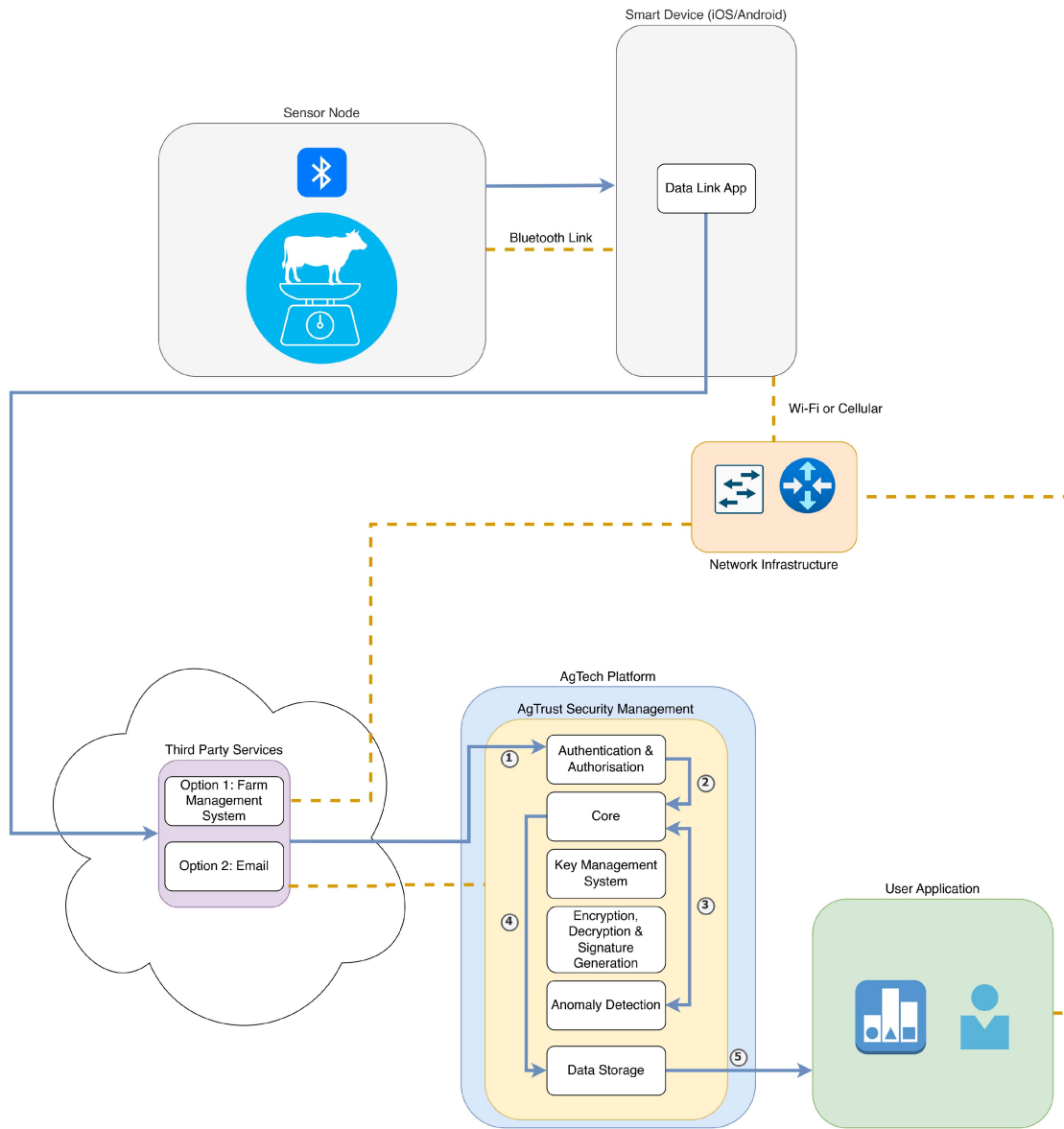


Fig. 13. Case study 3: Proprietary livestock sensor AgTrust data flow.

- 5) User applications can now access the data through secure APIs.

D. Scenario 4: AgTrust With Soil Moisture Sensor Node Based on Nonproprietary Heavily Resource-Constrained Hardware and Software

This implementation case study is similar in nature to the first case study presented. The developer has the ability to install custom software and firmware. This example presents a soil moisture, temperature, and electric current sensor that is connected to the farm network using Wi-Fi. This section outlines the required hardware and software components. It also presents a diagram that illustrates the data flow that a sensor reading takes using the AgTrust framework throughout its lifecycle, from collection, to storage and use.

The sensor node is based around an Arduino Nano with an 8-bit ATmega328 microcontroller. This device is heavily resource constrained having only 32 KB of flash memory, 2 KB of RAM with a 16 MHz clock speed [29]. The Arduino Nano is connected to a Sensecap S-Temp&VWC&EC-02B soil moisture, temperature, and EC sensor [26]. The sensor uses the MODBUS-RTU RS485 protocol [27] to communicate with the MCU. This requires translation as the Arduino cannot directly connect to the sensor. The sensor is connected to the Arduino via a MAX485 TTL UART to RS485 converter module. To enable connectivity, an HC-05 Bluetooth module is used.

Fig. 14 shows the network infrastructure required. Network links are represented by the orange dashed line. The data flow that is associated with the AgTrust framework is represented by the purple arrows. In this case study, the sensor has been

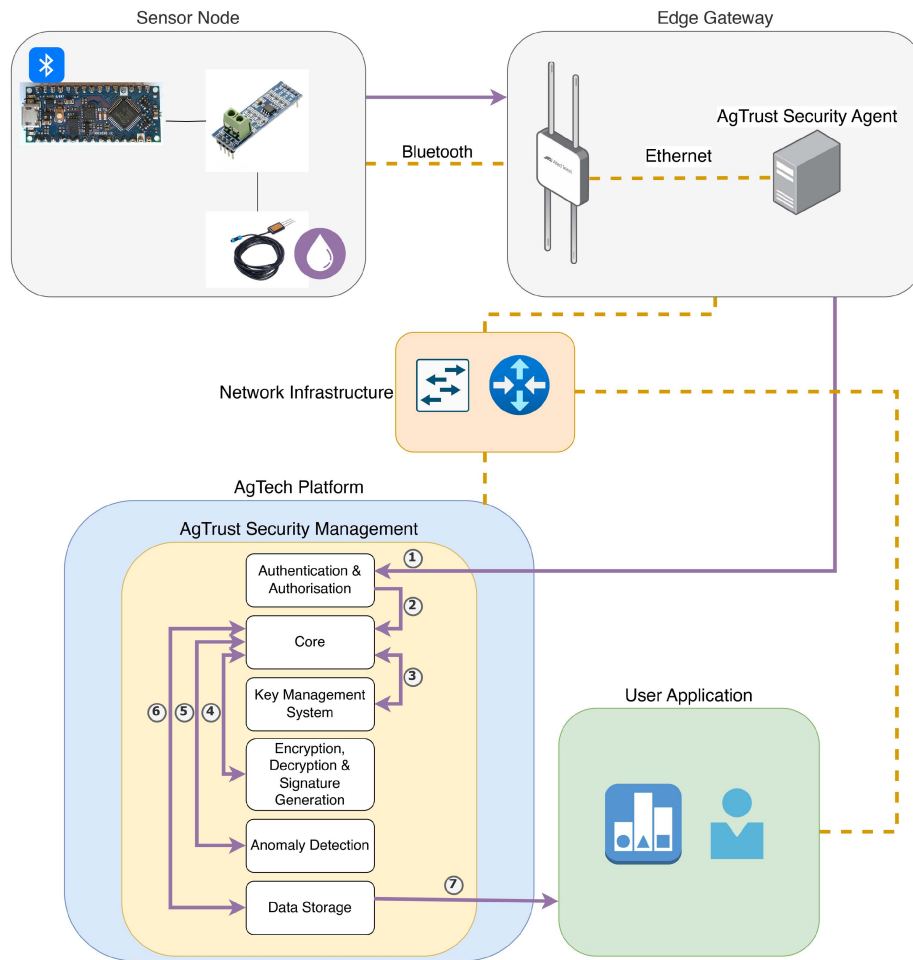


Fig. 14. Case study 4: resource-constrained soil moisture, temperature, and EC sensor AgTrust data flow.

provisioned with a set of keys to use for encryption. This was generated by the authentication and authorization module in the AgTrust framework and distributed to the AgTrust security agent within the edge gateway for distribution to the sensor node. These keys are used by the sensor node and the edge gateway to secure the communication channel. Once the data are received by the AgTrust security management platform, each stage of the data flow is numbered.

- 1) The transmission is received by the authentication and authorization module from the edge gateway. The authentication and authorization module ensures that the gateway is authentic, and has the appropriate authorization to provide the sensor data. It also ensures that the transmission is authentic.
- 2) Once the authentication and authorization module is satisfied of the authenticity of the sender and the message, the data are forwarded to the core module.
- 3) The transmission is received by the core module from the authentication and authorization module. The core is responsible for managing and coordinating all functionality related to the AgTrust security management system. A key is required to decrypt the sensor node readings. The

core module requests the appropriate key from the key management system.

- 4) The core passes the sensor data and the key to the encryption, decryption, and signature generation module. This module decrypts the sensor data and transmits it back to the core module.
- 5) The core module then forwards the data to the anomaly detection module. Anomaly detection processes are applied to the sensor data using previously trained ML models. These data are also used as input to further improve and train the models. Once the anomaly detection module is satisfied that the data is trustworthy, it is forwarded back to the core module.
- 6) The core module then forwards the data to the data storage module to be encrypted and saved.
- 7) User applications can now access the data through secure APIs.

VII. QUANTUM SAFETY AND HIGH-ENTROPY KEYS

The strength of any encryption method is partly determined by the randomness of the numbers used in the generation of

TABLE I
SUMMARY OF AGTRUST FEATURES, BENEFITS, AND COMPARISONS WITH EXISTING ARCHITECTURES

Features	Benefits	Existing references [7–11]
The proposed AgTrust framework ensures that the data used to make critical farming decisions is trustworthy and of high quality.	The framework provides a better structure that ensures data is validated for trustworthiness at the source, at the edge, and in the cloud.	This feature is not included in the Intel [7] and Cisco [8] models.
Data trustworthiness is constantly assessed, from the point of collection through to transmission, storage, and use.	This is a cohesive solution. Other approaches consider data security within the infrastructure but do not consider data trustworthiness at the point of collection (or beyond).	This feature is not included in the Intel [7], Cisco [8], Microsoft Azure [9], and IBM [10], [11] models.
Our proposed framework is not limited to agricultural industries. It can be applied to other IoT or smart-device-based networks.	The modules providing various functions can be adapted to ensure a future-proof design for a new purpose.	This feature is limited in the Intel [7], Cisco [8], Microsoft Azure [9], and IBM [10], [11] models.
The AgTrust framework is a modular design. It is not a vendor-specific solution. It is designed for cryptographic agility to meet a rapidly evolving threat landscape.	This allows flexibility and the concurrent use of different security mechanisms to ensure data trustworthiness. It provides more context around the security processes than the models in a vendor-agnostic manner.	This feature is limited in the Intel [7], Cisco [8], Microsoft Azure [9], and IBM [10], [11] models.

the keys used to protect the data [30]. This is true for both traditional and quantum-safe cryptographic techniques. Most current systems use pseudorandom number generators, which use seed values to generate bits. This type of random value generation is low-entropy and will be unsuitable for quantum-safe applications [31].

Using a high-entropy source for random numbers is not only essential for quantum safety, it is also important in safeguarding and future proofing the security of classical encryption algorithms. The AgTrust framework relies on integration of high-entropy bit generation using commercial products.

The QuintessenceLabs [32] qStream quantum random number generator (QRNG) [33] is a hardware device that can provide quantum source true random numbers. This device can be implemented in existing infrastructure in the form of a PCIe card, or deployed as a standalone network appliance. Also, the Trusted Security Foundation Enterprise Key and Policy Manager [34] is another example of such a solution, a comprehensive key-management appliance with the true random number generation and hardware security modules. Note that the AgTrust framework is modular. Alternative solutions can be substituted to ensure high-entropy randomness.

In an agricultural context, all the various sensor devices, gateways, and other related systems can make use of the QRNG for the purposes of supporting the cryptographic operations. In practice, a QRNG device could be placed in a centralized location within the data center and issue high-entropy bits on demand to clients and/or gateway devices. Another option is to follow the centralized model with a cache in each gateway. The gateway could store these high-entropy bits within a secure enclave and distribute to sensor end nodes when required. In situations where full centralization is not possible, a distributed design may be implemented with local gateways having the QRNG locally available. The high-entropy bits could then be provided to sensor nodes as required.

VIII. DISCUSSION AND COMPARISON

The reviewed IoT architecture models [7], [8], [9], [10], [11] in Section III are commonly aligned with a particular vendor's product range. The security components of these reference models are often not well-defined or explained and may not be available for scrutiny. The way that security mechanisms should be implemented is not clear. This increases the complexity of coordinating operations of products from multiple vendors in a secure way.

The AgTrust framework provides more context around the security processes than the reference models in a vendor-agnostic manner (refer to Table I for comparisons). Unlike the Intel IoT system architecture [7] and the Cisco IoT security reference model [8], in our framework authentication, authorization, and secure storage are clearly identified and defined, and the exact mechanisms that should be placed and implemented are clearly specified. Comparing with the Microsoft Azure IoT reference architecture [9] and the IBM IoT reference architecture [10], [11], our framework is not a vendor-specific solution.

Applying the AgTrust framework ensures that the data used to make critical farming decisions is trustworthy and of high quality. Data trustworthiness is constantly assessed, from the point of collection through to transmission, storage, and use. It provides a structure that ensures data is validated for trustworthiness at the source, at the edge, and in the cloud. Provisions are also included to ensure data at rest is encrypted to mitigate against potential data breaches. Other architectures consider data security within the infrastructure but do not consider data trustworthiness at the point of collection (or beyond). It is a necessary consideration.

The AgTrust framework is a modular design. This allows flexibility and the concurrent use of different security mechanisms to ensure data trustworthiness. The framework specifies the functionality required across various layers and services and coordinates interactions between functions. The modules

providing various functions can be adapted to ensure a future-proof design. The core module within the AgTrust security management system is responsible for coordinating all activities between different modules and agents. All communication between modules is via the core. Table I summarizes the differences between this invention and what was done in the past.

IX. CONCLUSION AND FUTURE DIRECTION

AgTech solutions involve many interconnected systems, each with a range of security vulnerabilities that may impact data trustworthiness. This article introduced a novel framework “AgTrust,” a modular design. This allows flexibility and the concurrent use of different security mechanisms to ensure data trustworthiness.

AgTrust framework provides a cohesive solution for farming ecosystems, including assurances of data integrity, data quality, and data provenance supported by anomaly detection and quantum-safe encryption, authentication, and the underlying key-management mechanisms.

Four case study examples related to the implementation of the AgTrust framework in real-world scenarios are presented and discussed. The AgTrust features and benefits are compared with well-known architectures. This demonstrated the effectiveness of our proposal, where the modular design of the framework enables security component choices appropriate for different farming contexts.

Our framework is not limited to agricultural industries. It can be applied to other IoT or smart-device-based networks. The modules providing various functions can be adapted for cryptographic agility to future proof the framework.

Applying the framework in real-world farming ecosystems will allow us to monitor the operational aspects and explore integration with other third-party services. This potential avenue for future research allows us to find possible rooms for improvements.

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